Diffusion-Based Stock Scenario Generator

with a Simple Option-Pricing Layer

Deep Learning Course Project

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Code, slides & figures:
github.com/NDamirov/market-diffusion-simulator

Outline

Why Synthetic Market Scenarios?

- Risk management requires rare stress events that do not appear often in historical data
- Parametric models (Black-Scholes, GBM) assume Gaussian returns — markets are fat-tailed
- Deep generative models can learn complex, non-Gaussian return dynamics directly from data

Inspiration:
Multi-Asset Spot and Option Market Simulation (Wiese et al., 2021) — deep generative simulators for joint spot + option markets

Project Scope

Educational prototype — what we build

1. Download daily stock prices for 5 liquid assets
2. Convert to daily log-return windows of length 20
3. Train a **DDPM-style denoising diffusion model** on multi-asset windows
4. Generate synthetic return scenarios and compare vs. a **Gaussian baseline**
5. Extend to a **conditional DDPM** for calm / normal / stress market regimes
6. Add a **Black-Scholes pricing layer** to translate scenarios into call option prices

Data & Preprocessing

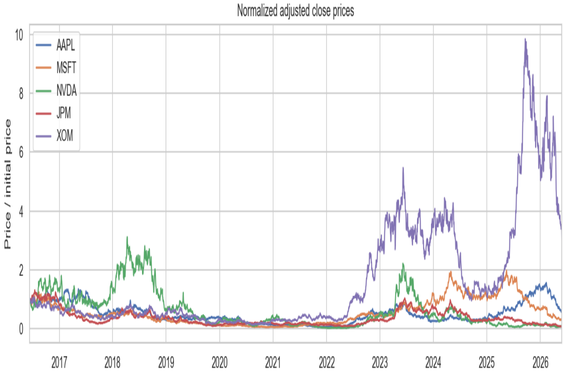
Assets: AAPL, MSFT, NVDA, JPM, XOM **Period:** 2014 – 2025

Pipeline:

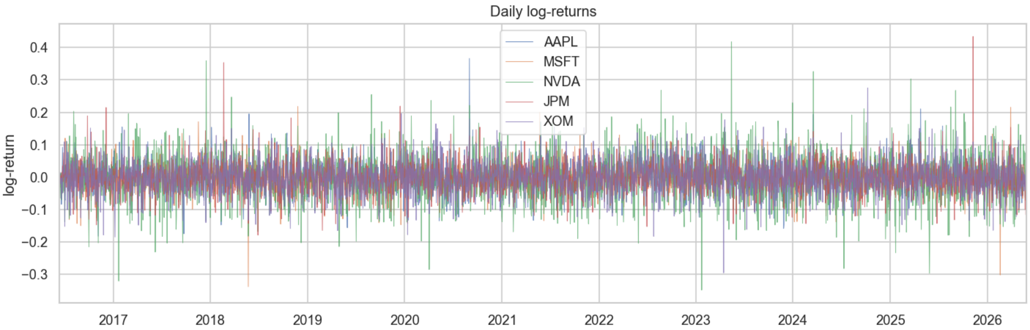
1. Daily **log-returns**:
   $r_t = \ln S_t - \ln S_{t-1}$
2. Train / test split **80 / 20**
3. **Standardise** with train mean & std
4. Rolling **20-day windows**:

$$X_i = (r_i, \ldots, r_{i+19}) \in \mathbb{R}^{20\times 5}$$

5. Clip returns at $\pm 35\%$



Daily Log-Returns



- Volatility clearly clusters in time (ARCH effect)
- NVDA exhibits the largest individual swings ($\pm 20\%$ daily in extreme events)

Denoising Diffusion Probabilistic Model

Forward process — gradually add noise:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\left(\sqrt{1-\beta_t}\, x_{t-1},\ \beta_t I\right)$$

Training objective — predict the noise:

$$\mathcal{L} = \mathbb{E}_{x_0,t,\epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2\right]$$

Reverse process — iterative denoising from $x_T \sim \mathcal{N}(0, I)$

Schedule: linear $\beta_t \in [10^{-4}, 0.02]$, $\quad T=100$ steps

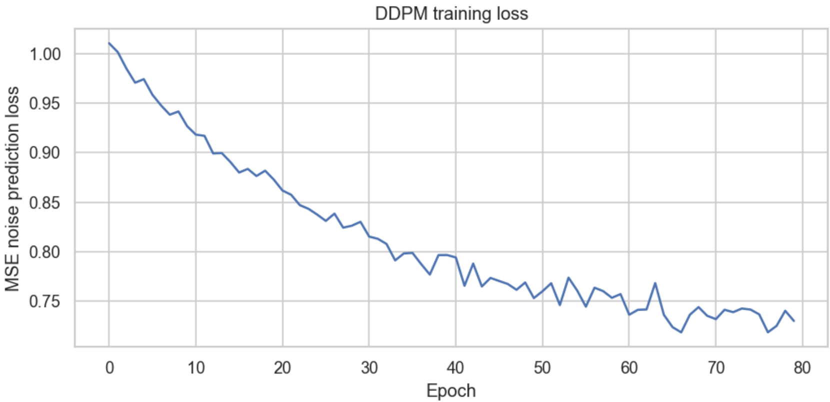
Network: DenoiseMLP

Input: window $\mathbb{R}^{100}$ + time emb. $\mathbb{R}^{64}$

Layer	Size
Linear + SiLU + LN	$164 \to 256$
Linear + SiLU + LN	$256 \to 256$
Linear + SiLU	$256 \to 256$
Linear (output)	$256 \to 100$

Optimiser: AdamW, lr $= 2 \times 10^{-4}$
Epochs: 500 (GPU) / 250 (CPU)

Training Curve



- Loss decreases steadily — model learns to predict noise in return windows
- MSE $\approx$ 0.62 at convergence; gradient clipping (max norm 1.0) stabilises training

Gaussian Baseline

Multivariate Gaussian fitted to flattened windows

$$\hat{x} \sim \mathcal{N}(\hat{\mu}, \hat{\Sigma}), \quad \hat{\mu} = \frac{1}{N}\sum_i x_i, \quad \hat{\Sigma} = \mathrm{Cov}(x_1, \ldots, x_N)$$

Strengths vs. per-day Gaussian:

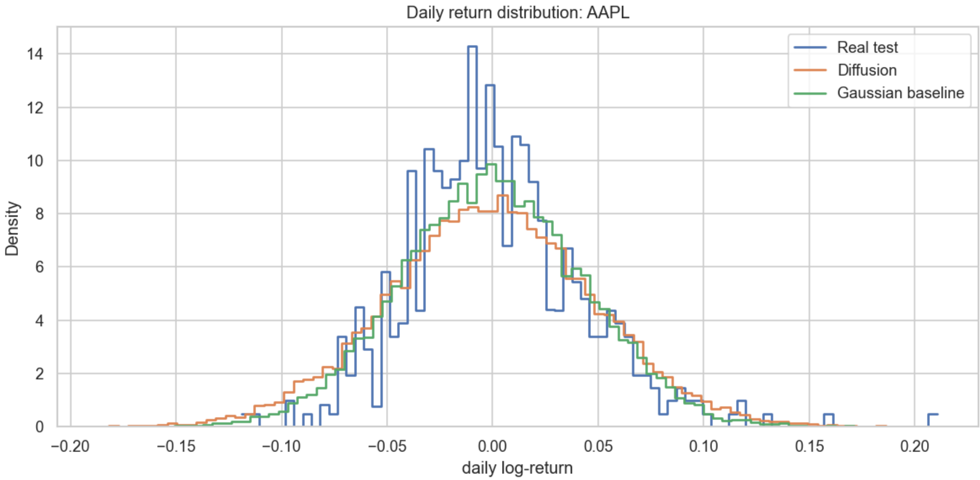
- Captures *linear* cross-asset and cross-time covariance

Weaknesses:

- Cannot represent fat tails or skewness
- Cannot model volatility clustering
- All marginals are Gaussian by construction

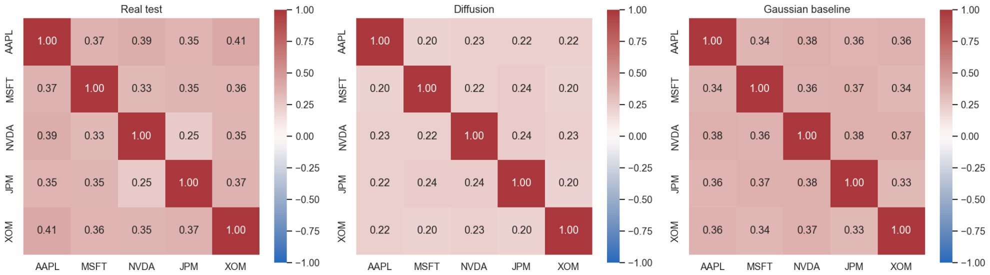
Generation: 3 000 windows sampled from each model for all evaluations.

Return Distribution: AAPL



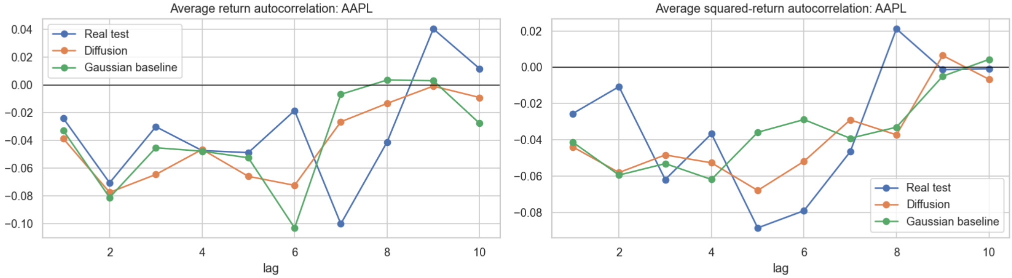
- Real returns: excess kurtosis ≈ 11.6, heavy tails
- Both models produce lighter tails; diffusion slightly heavier than Gaussian

Correlation Structure



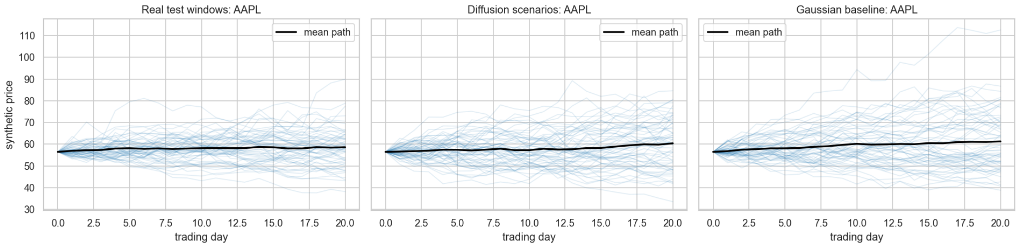
- Diffusion Frobenius error vs. real ≈ 0.27; Gaussian baseline ≈ 0.35

Stylized Facts: Autocorrelation



- **Left** — return ACF near-zero for all (efficient markets reproduced)
- **Right** — squared-return ACF: real data has positive lags (ARCH); diffusion partially captures this, Gaussian cannot

Synthetic Price Paths



- 60 paths per panel, mean path in black; $S_{t+1} = S_t \exp(r_{t+1})$
- Diffusion fan-out more irregular and realistic than Gaussian

Portfolio VaR and CVaR

Equal-weight portfolio, 5% tail, daily returns:

Dataset	VaR_{95} loss	CVaR_{95} loss
Real test	0.0170	0.0276
Diffusion	0.0175	0.0228
Gaussian baseline	0.0239	0.0301

- Diffusion VaR matches real data well; Gaussian **overestimates** tail losses
- Diffusion CVaR slightly **underestimated** — extreme events still missed

Conditional DDPM: Idea & Regime Labels

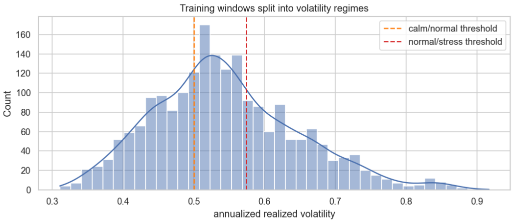
Goal: controllable generation — produce scenarios conditioned on a market regime y.

$$\epsilon_\theta(x_t,\, t,\, y), \quad y \in \{\text{calm, normal, stress}\}$$

Labels from each window's annualized portfolio realized volatility:

- **Calm** — bottom third
- **Normal** — middle third
- **Stress** — top third

Regime label passed as a **learned embedding** of size 32.



Classifier-Free Guidance

Training

Regime label y is randomly replaced by `unknown` with probability 15%.
This trains the model to work both conditionally and unconditionally.

Sampling

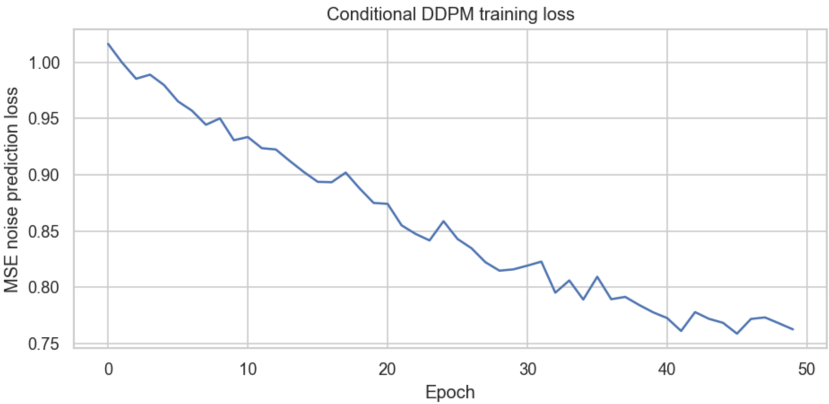
Noise prediction is blended:

$$\hat{\epsilon} = \epsilon_{\text{unc}} + s\left(\epsilon_{\text{cond}} - \epsilon_{\text{unc}}\right), \qquad s = 1.5$$

Higher s pushes the sample further towards the requested regime.

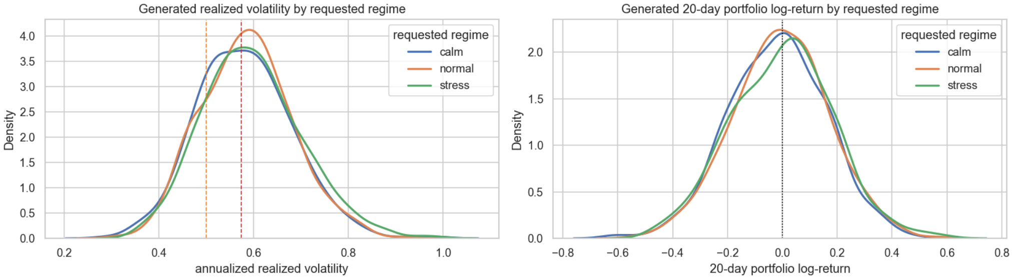
Epochs: 250 (GPU) / 120 (CPU), same MLP backbone as unconditional model.

Conditional Training Loss



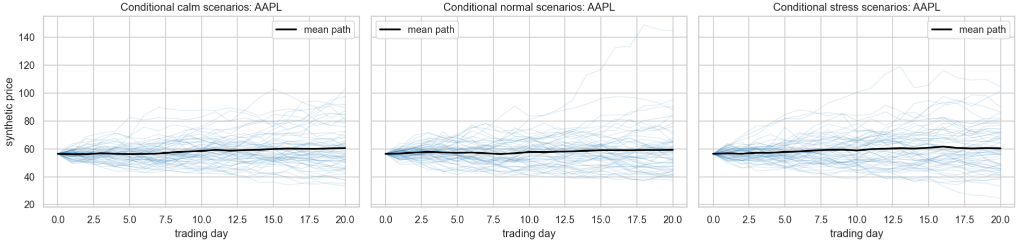
- Loss converges similarly to the unconditional model
- Regime embedding adds negligible parameter overhead

Conditional: Volatility & Return Distributions



- **Left**: stress regime has noticeably higher realized volatility
- **Right**: stress regime has heavier left tail in 20-day portfolio return

Conditional Price Paths by Regime



- **Calm**: tight fan — **Normal**: moderate spread — **Stress**: wide fan with large drawdowns

Conditional Regime Risk Summary

Requested regime	Mean vol	20d VaR_{95}	20d $CVaR_{95}$	Share stress
Calm	0.159	0.035	0.056	1.6%
Normal	0.178	0.047	0.069	7.5%
Stress	0.209	0.085	0.112	29.3%

- Risk metrics monotonically increase — conditioning clearly works
- Stress VaR (8.5%) is **2.4**× higher than calm (3.5%)

Black-Scholes Option Layer: Idea

Goal: translate each generated 20-day stock scenario into a synthetic European call price.

For each window:

1. Compute terminal price $S_{\mathrm{end}} = S_0 \exp\left(\sum_{t=1}^{20} r_t\right)$
2. Estimate scenario realized vol σ_{sc} from the window
3. Price call at horizon with Black-Scholes:

$$C = S\,N(d_1) - K\,e^{-rT}\,N(d_2)$$

$$d_1 = \frac{\ln(S/K) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}$$

Important: this is a **transparent derivative pricing layer**, not a learned option surface.

Black-Scholes Layer: Setup & Portfolio

Option parameters

- Asset: AAPL
- Strike: ATM ($K = S_0$)
- Maturity: 30 trading days
- Risk-free rate: 4%
- C_0 priced with trailing 252-day realized vol

Combined portfolio

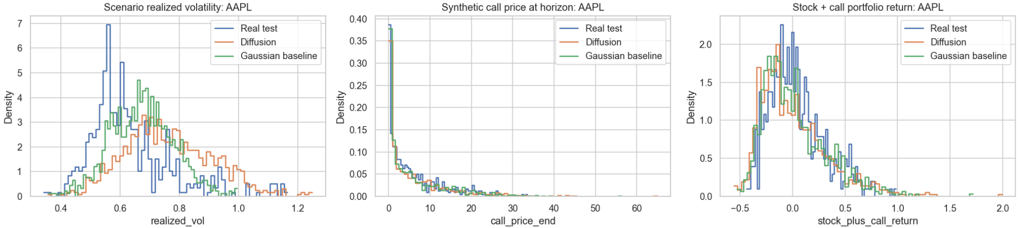
Hold one share + one call option.

$$\text{PnL} = (S_{\text{end}} - S_0) + (C_{\text{end}} - C_0)$$

$$\text{Return} = \frac{\text{PnL}}{S_0 + C_0}$$

- Allows comparison of tail risk metrics across real / diffusion / Gaussian scenarios

Option Layer Results



- Diffusion vol matches real test more closely than Gaussian
- 20d Stock+Call $CVaR_{95}$: Real 0.153, Diffusion 0.147, Gaussian 0.162

Summary: What Was Built & Key Results

What was built:

- DDPM trained on 20-day multi-asset return windows (AAPL, MSFT, NVDA, JPM, XOM)
- Comparison against a multivariate Gaussian baseline
- Conditional DDPM with regime embeddings + classifier-free guidance
- Black-Scholes derivative pricing layer on generated paths

Key results:

- Diffusion VaR closer to real data than Gaussian
- Correlation structure better preserved (lower Frobenius error)
- Volatility clustering partially captured by diffusion, not by Gaussian
- Regime conditioning works: $2.4\times$ higher VaR for stress vs. calm

Limitations & Future Work

Current limitations

- **Daily** frequency only — no intraday dynamics
- Short windows (20 days) — long-range macro effects not captured
- Marginal distributions still near-Gaussian (fat tails not fully reproduced)
- No true option surface learning; no arbitrage-free constraints
- No order book, no macro variables, small dataset ($\approx$3000 days)

Possible extensions:

- Score-based diffusion or flow matching for better tail modeling
- Conditioning on macro signals or option-implied vol
- Longer windows with transformer-based noise predictor

Conclusion

Project in one paragraph

We implemented a compact DDPM-style diffusion model for 20-day multi-asset stock return windows and compared it with a multivariate Gaussian baseline. The model generates synthetic short-term market scenarios evaluated by distributional statistics, cross-asset correlations, autocorrelation patterns, and portfolio risk measures. We extended the base model to a **conditional DDPM with learned volatility-regime embeddings**, enabling controllable generation of calm, normal, and stress scenarios. A simple Black-Scholes pricing layer translates generated paths into synthetic European call option prices and stock-plus-option PnL.